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Final Project Reflection

While replicating the 2D image I selected in module one, I used various different meshes such as boxes, cylinders, cones, a tapered cylinder, and a half torus. I chose these shapes because they best replicate the objects displayed in the 2D scene. The objects that I replicated are the following, a computer monitor, a pencil holder, pencils, a stack of books, and a desk. I chose to replicate these objects because they use various meshes and display that I have the ability to replicate objects using a single mesh and while using multiple meshes. To achieve the required functionality, I implemented transformations such as scaling, rotation, and translation to position each object within the scene. I also incorporated textures, materials, and lighting to accurately represent the scene.

To navigate the scene, the keys Q, W, E, A, S, and D can be used. The keys Q and E allow the user to navigate the scene vertically up and down while the other keys W, A, S, and D allow the user to navigate the scene forward, back, left, and right. Additionally, I added a free-fly camera to seamlessly navigate the scene. The mouse also assists the user in navigation. Allowing the user to steer the camera by looking in a direction while controlling the speed of the camera using the mouse scroll wheel.

Some custom functions within my code that make it modularized and organized are LoadSceneTextures(), DefineObjectMaterials(), and SetupSceneLights(). These are all functions in which I created organized code to separate major responsibilities within the program. The LoadSceneTextures() function handles loading and managing all textures used in the scene, DefineObjectMaterials() stores material properties for objects including ambient, diffuse, and specular values. Finally, SetupSceneLights() configures the lighting used within the scene.